NEU C64: Exploring the Brain: Introduction to Neuroscience

Summer 2026 | Tu/Th 10am-12pm | 100 Genetics & Plant Bio

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Teaching Staff

Instructor: Tara Raam, Ph.D., tararaam@berkeley.edu
GSI: Marley Lovemann-Brown, mbloveman@berkeley.edu

Course Information

Lectures	Tues/Thurs 10am – 12pm, 100 Genetics & Plant Bio
Discussion Sections	Friday 10am-12pm & 1pm-3pm
Final Exam	Thursday August 13th, 10am – 12pm
Prerequisites	High school chemistry & biology
Course Website	tararaam.github.io/neu-c64
bCourses	https://bcourses.berkeley.edu/courses/1555229

Office Hours:

Staff	**Date & Time**	**Location**
Professor Raam	Wed. 9am-10am	via Zoom, link to be provided
Marley Lovemann-Brown, GSI	TBD	TBD

Important Dates:

Start of Instruction	June 23rd
Last day to add or drop	July 2nd
Last day to change grade options	July 31st
Last day of class	August 13th

Important Note: *Students will not receive credit for NEU C64 after completing NEU C61, MCELLBI 104, CHEM C130, MCELLBI 110, MCELLBI C130, MCELLBI 136, MCELLBI 160, or INTEGBI 132.*

Course Description

This course will introduce lower division undergraduates to the fundamentals of neuroscience. The first part of the course covers basic membrane properties, synapses, action potentials, chemical and electrical synaptic interactions, receptor potentials, and receptor proteins. The second part of the course covers networks in invertebrates, memory and learning behavior, modulation, vertebrate brain and spinal cord, retina, visual cortex architecture, hierarchy, development, and higher cortical centers.

Learning Goals

By the end of this course, you will be able to…

1. **Explain the building blocks of nervous system function**, from neurons and glia to action potentials and synaptic transmission.

2. **Describe how the brain transforms sensation into perception and experience**, from the chemical senses, vision, and hearing to emotion, sleep, language, and memory.

3. **Understand how neuroscientists think and ask questions about the brain:** what makes a question scientifically tractable, how experimental approaches are matched to specific questions, and what each method can and cannot tell us.

4. **Use AI thoughtfully as a learning tool**, developing stonger judgment for when AI deepens your understanding and when it will shortcut it.

5. **Communicate neuroscience concepts accessibly to a broad audience** through written reflections and group presentations.

6. **Connect neuroscience to everyday life**, relating course concepts to health, education, and society.

Materials

Required Textbook: Neuroscience: Exploring the Brain 5th Ed. Mark F. Bear, Barry W. Connors, Michael A. Paradiso. ISBN: 1284286878

This book is available at the campus bookstore, however you may also be able to find it from other sources. An E-Book is also available online (CalCentral log-in required) for a maximum of 3 users at a time at this link.

All course materials will be provided via the course website. bCourses will be used for announcements, discussion boards, submitting assignments, and to view your grades.

Course Calendar

Week	Date	Topic	Reading Due	Take-Home Quizzes	Assignments	Group Project
1	6/23	Intro, History, Neuroanatomy	Ch 1 Ch 7: p194-203	Pre-course survey due Fri 11:59pm		
	6/25	Neurons & Glia	Ch 2			
2	6/30	Membrane Potential	Ch 3	Q1 due Sun 11:59pm	In-Class Reflection 1	Introduce Project + Form Groups
	7/2	Action Potential	Ch 4			
3	7/7	Synaptic Transmission, Neurotransmitters	Ch 5: p114-141 Ch 6: p154-173	Q2 due Sun 11:59pm	Homework 1 due Tue 11:59pm	
	7/9	Probing the Brain + Review	p 86, 90, 203-209, 705-712, 840, 843-844			
4	7/14	MIDTERM	--	Q3 due Sun 11:59pm		Checkpoint 1 due Friday at 11:59pm
	7/16	Chemical Senses	Ch 8			
5	7/21	Vision	Ch 9-10	--	Homework 2 due Tue 11:59pm	
	7/23	Hearing	Ch 11		In-Class Reflection 2	
6	7/28	Emotions	Ch 18	Q4 due Sun 11:59pm		Present in Discussion Section
	7/30	Sleep	Ch 19			
7	8/4	Language	Ch 20	Q5 due Sun 11:59pm		Present in Discussion Section
	8/6	Memory	Ch 24 Ch 25: p944-966		In-Class Reflection 3	
8	8/11	Topics of Interest + Review	--	Q6 due Sun 11:59pm	Homework 3 due Tue 11:59pm	
	8/13	FINAL EXAM	--			

Assignments

1. Homework Assignments: You will have 3 homework assignments in total, which will be focused on applying concepts you've learned in class. You will turn in your homework assignments on bCourses.

2. In-class Written Reflections: We will have 3 in-class written reflections on the dates noted on the schedule above. These will be designed to help you think more deeply about the applications and implications of what we have learned. In-class reflections can only be made up with a documented excused absence.

3. Weekly Take-Home Quizzes: There will be 6 take-home quizzes based on material from the readings and lectures. These will be completed and submitted via bCourses. As they are to be completed at home, they are open note and you can see them as low-stress ways to test your knowledge of course material.

4. Exams: We will have two exams – a midterm in Week 4, and a final exam in Week 8. These will be multiple choice scantron exams. We will hold review sessions before each exam. Reflections can only be made up with a documented excused absence.

5. Group Project: The group project explores an aspect of neuroscience that is broadly relevant to society and teaches it in an accessible format to a non-scientist audience. More information will be provided in week 2.

6. Discussion Section Participation: Discussion section attendance and participation are part of your grade, and you are strongly encouraged to attend.

Grading

Your final letter grade in this course will be based on a point accumulation system, rather than a percentage system.

Category	**Course Component**	**Points**
Misc.	Pre-course Survey	1
	Course Evaluation	1
Assignments	In-class Reflections (3 total)	15
	Homework Assignments (3 total)	12
	Take-home Quizzes (6 total)	6
	Group Project	15
Exams	Midterm	20
	Final	25
Participation	Section Participation	10
	TOTAL:	**105**

As you may notice, there are a **total of 105 possible points** you can earn in this course. The 5 extra points offer flexibility for unprecedented situations that may come up, and **take the emphasis off grades, so that you can focus on learning** and enjoying the course material.

How you choose to earn points towards your final grade is entirely up to you! For example, you can choose to skip reading quizzes, not turn in an assignment, or participate less in discussion section. Nothing is strictly required and everything is optional – your learning is your own responsibility! 😊

Final letter grades will be assigned according to the following grading scheme:

Points Earned	Letter Grade
>97	A+
93-96	A
90-92	A-
87-89	B+
83-86	B
80-82	B-
77-79	C+
73-76	C
70-72	C-
60-69	D
<59	F

PLEASE NOTE: Because there are ample opportunities to earn points in this course, we will not be offering extra credit, make-up assignments for unexcused absences, or rounding borderline grades – this is what the extra 5 points are for! Please do not email course staff with these requests.

Policies & FAQ

Attendance Policy

We aim to make lecture as useful as possible by doing interactive exercises and activities in addition to lecture – hopefully you find it worthwhile to attend. While we won't be taking attendance during lecture, there will be 3 in-class written assignments that earn you points toward your grade. Make-ups for these assignments will not be offered unless you have a documented medical absence.

Section attendance and participation is highly encouraged and earns you 10 points toward your grade.

Late Assignments Policy

Homework assignments and take-home quizzes can be turned in a maximum of 5 days late, but will incur a 10% deduction per late day (the max deduction is 50%).

Group project checkpoints and presentations must be turned in on time to receive credit.

What happens if I need to miss an exam?

If you know you need to miss an exam for a legitimate reason (illness, emergency, etc.), please contact Professor Raam ASAP to schedule a make-up exam. A documented excuse will be required. Make-up exams for unexcused absences will not be permitted.

What should I do if I need learning accommodations?

Please make sure to have your accommodations noted with the Disabled Students Program. If your accommodations enable you to take exams under specific conditions, please make sure to contact the DSP office one week in advance of each exam to ensure you will be able to take your exam in the DSP room. Contacting DSP on time is your responsibility, and course staff do not have control over DSP testing space if the deadline is missed.

What is the academic integrity policy for this course?

Academic dishonesty is forbidden and is reportable to the Center for Student Conduct. If you are struggling with course material, please reach out to me or the GSI so we can help you. Resorting to cheating is unfair to other students, will create bad habits in yourself, and leave you unprepared to tackle future challenges in school and in work.

The following are considered academic dishonesty: plagiarism, copying answers from someone else during an exam, talking to another student during an exam, using your phone or other technology during closed-book assessments, faking your or another student's attendance, copying AI-generated material and submitting it as your own work, altering answers on a graded assessment and then asking for a re-grade.

Assignments in which academic dishonesty are discovered will be recorded as a zero.

Can I use ChatGPT or other AI tools to help me in this course?

Yes, you are welcome to use AI to help you learn in this course! I regularly use AI to help me learn new things, and I find it to be a wonderful tool. In fact, I have custom built interactive tools to help you learn challenging course concepts with the help of an AI tutor, and I hope you make use of them to help you learn.

HOWEVER, there are helpful ways to use AI that will enhance your learning, and there are also detrimental ways to use AI that will cause brain rot. **Chronic, mindless use of AI will leave you unprepared for your education and, ultimately, life.** You should always be aware that AI tools are imperfect, can sometimes hallucinate (create false or incorrect answers to your questions), and are prone to bias. Anytime you use AI to help you learn, you should be conscious of these caveats and double check what you are learning from other sources. You are ultimately responsible for your own learning, and fact-checking AI output is an important element of that.

Throughout the course, I'll share examples of productive ways to use AI, such as helping you clarify difficult concepts, pushing you to consider new points of view, and getting feedback on your work. However, **copying AI-generated output and submitting it as your own work is strictly prohibited**. Not only is this very obvious, but it also undermines your opportunity to learn.

How do I provide feedback about the course?

This is my first time teaching this course, so I am certainly open to any feedback you may have about course topics, lectures, specific assignments, and the interactive AI tools I've built for you. You can always talk to me or the GSI during our office hours, or write to us via email, but if you prefer an anonymous route, you can use this feedback form at any point in the semester.